

AI Assistant – Project Documentation

1. Project Overview

The AI Assistant is a web-based application built using Python and Flask. It utilizes prompt engineering techniques and integrates with the OpenRouter API to perform three main tasks:

- **Answer Questions:** Provides factual responses to user queries.
- **Summarize Text:** Produces concise summaries from longer text inputs.
- **Generate Creative Content:** Creates stories, poems, or ideas based on prompts.

The application includes a feedback mechanism where users can indicate whether a response was helpful. Feedback is stored in a log file for later analysis.

2. Environment Setup

Step 1: Create a Virtual Environment

```
python -m venv venv
```

```
venv\Scripts\activate (Windows)
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

3. Application Architecture

User → Web Interface → Prompt Engine → OpenRouter API → Response → Feedback Logging

4. Web Application Code

Snippet 1 – Prompt Templates

```
if request.method == 'POST':
    prompt_type = request.form.get('task')
    user_input = request.form.get('user_input')
    feedback = request.form.get('feedback', '')

    if prompt_type == 'question':
        prompt = user_input
    elif prompt_type == 'summary':
        prompt = f"Summarize this text briefly:\n{user_input}"
    elif prompt_type == 'creative':
        prompt = f"Create something based on this idea: {user_input}"
    else:
        prompt = user_input

    result = get_response(prompt)
```

Snippet 2 – OpenRouter API Integration

```
#OpenRouter API Setup
# 🗝️ API key and model setup
import os
from dotenv import load_dotenv

# 🌱 Load environment variables from .env file
load_dotenv()

# 🗝️ Securely get the API key from .env
API_KEY = os.getenv("OPENROUTER_API_KEY")

MODEL = "openai/gpt-3.5-turbo"
```

Snippet 3 – Main Route

```
@app.route('/', methods=['GET', 'POST'])
def index():
    result = ""
    prompt_type = ""
    user_input = ""
    feedback = ""

    if request.method == 'POST':
        prompt_type = request.form.get('task')
        user_input = request.form.get('user_input')
        feedback = request.form.get('feedback', '')

        if prompt_type == 'question':
            prompt = user_input
        elif prompt_type == 'summary':
            prompt = f"Summarize this text briefly:\n{user_input}"
        elif prompt_type == 'creative':
            prompt = f"Create something based on this idea: {user_input}"
        else:
            prompt = user_input

        result = get_response(prompt)
```

Snippet 4 – Feedback Route

```
# 🍷 Log feedback
if feedback:
    with open("feedback.txt", "a", encoding="utf-8") as f:
        f.write(f"Prompt: {prompt}\nFeedback: {feedback}\n\n")
```

5. HTML Template (templates/index.html)

Snippet 1 – HTML Head & Styling

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Your AI Assistant</title>
  <style>
    :root {
      --primary: #0074cc;
      --primary-dark: #005a99;
      --accent: #3399ff;
      --background: #eaf4fc;
      --text: #333;
      --white: #ffffff;
      --shadow: rgba(0, 0, 0, 0.1);
    }

    body {
      font-family: 'Segoe UI', Tahoma, sans-serif;
      background-color: var(--background);
      color: var(--text);
      margin: 0;
      padding: 40px 20px;
    }

    .container {
      max-width: 800px;
      margin: auto;
      background-color: var(--white);
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0 10px 25px var(--shadow);
    }

    h1 {
      text-align: center;
      color: var(--primary);
      margin-bottom: 30px;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>Your AI Assistant</h1>
  </div>
</body>
</html>
```

```
label {
  font-weight: 600;
  margin-top: 20px;
  display: block;
  color: var(--primary-dark);
}

select, textarea, input[type="submit"] {
  width: 100%;
  padding: 12px;
  font-size: 16px;
  margin-top: 8px;
  margin-bottom: 20px;
  border-radius: 8px;
  border: 1px solid #ccc;
  box-sizing: border-box;
}

textarea::placeholder {
  color: #aaa;
}

select:focus, textarea:focus {
  border-color: var(--primary);
  outline: none;
}

input[type="submit"] {
  background-color: var(--primary);
  color: white;
  font-weight: bold;
  border: none;
  cursor: pointer;
  transition: background-color 0.3s ease;
}

input[type="submit"]:hover {
  background-color: var(--primary-dark);
}
```



```

    .result {
      background-color: var(--white);
      border-left: 6px solid var(--accent);
      padding: 20px;
      margin-top: 30px;
      border-radius: 10px;
      box-shadow: 0 4px 12px var(--shadow);
    }

    .result h3 {
      margin-top: 0;
      color: var(--primary-dark);
    }

    footer {
      text-align: center;
      margin-top: 50px;
      font-size: 14px;
      color: #666;
    }
  }
</style>
</head>

```

Snippet 2 – Main Body & Input Form

```

<body>
  <div class="container">
    <h1>👋 Welcome to Your AI Assistant</h1>

    <!-- Task Selection and Input -->
    <form method="POST">
      <label for="task">What do you need help with today?</label>
      <select name="task" id="task" required onchange="clearInputAndResponse()">
        <option value="question">🗣️ Ask a question</option>
        <option value="summary">📄 Summarize the text</option>
        <option value="creative">💡 Generate creative content</option>
      </select>

      <label for="user_input">Your prompt or text:</label>
      <textarea name="user_input" id="user_input" rows="6" placeholder="Type or paste your content here..." required>{{ user_input }}</textarea>

      <input type="submit" value="👉 Ask the Assistant">
    </form>
  </div>
</body>

```

Snippet 3 – Response Display & Feedback Form

```

<!-- AI Response Output -->
<div id="response-block">
  {% if result %}
    <div class="result">
      <h3>🧠 Here's what I came up with:</h3>
      <p>{{ result }}</p>
    </div>

    <!-- Feedback Form -->
    <form method="POST">
      <input type="hidden" name="task" value="{{ task }}">
      <input type="hidden" name="user_input" value="{{ user_input }}">
      <input type="hidden" name="result" value="{{ result }}">

      <label for="feedback">Was this helpful to you?</label>
      <select name="feedback" id="feedback" required>
        <option value="">-- Choose an option --</option>
        <option value="yes">✅ Yes, it was helpful</option>
        <option value="no">❌ Not really</option>
      </select>

      <input type="submit" value="🗨️ Submit Feedback">
    </form>
  {% endif %}
</div>
</div>

```

6. Example Inputs & Outputs

Input for Function Generate Creative Content:

What do you need help with today?

🗨️ Ask a question

Your prompt or text:

who won the first noble prize for India

🧠 Ask the Assistant

🧠 Here's what I came up with:

Rabindranath Tagore was the first Indian to win the Nobel Prize. He was awarded the Nobel Prize in Literature in 1913 for his book of poems, "Gitanjali".

Was this helpful to you?

✅ Yes, it was helpful

🗨️ Submit Feedback

What do you need help with today?

🔍 Ask a question

Your prompt or text:

Global warming results from the increasing concentration of carbon dioxide (CO2) and other greenhouse gases that trap heat in the earth's atmosphere. This trapped energy comes from incoming solar radiation absorbed by the earth's surface and re-radiated back into the atmosphere as infrared energy.

Though natural cycles and fluctuations have caused the planet's climate to change continually over the last 800,000 years, it is human activity that has driven our

🔥 Ask the Assistant

💡 Here's what I came up with:

Global warming is caused by an increase in greenhouse gases, particularly carbon dioxide, which trap heat in the earth's atmosphere. Human activity, especially the burning of fossil fuels, is the main driver of global warming. The largest sources of greenhouse gases in the United States are transportation, electricity production, and industrial activity.

What do you need help with today?

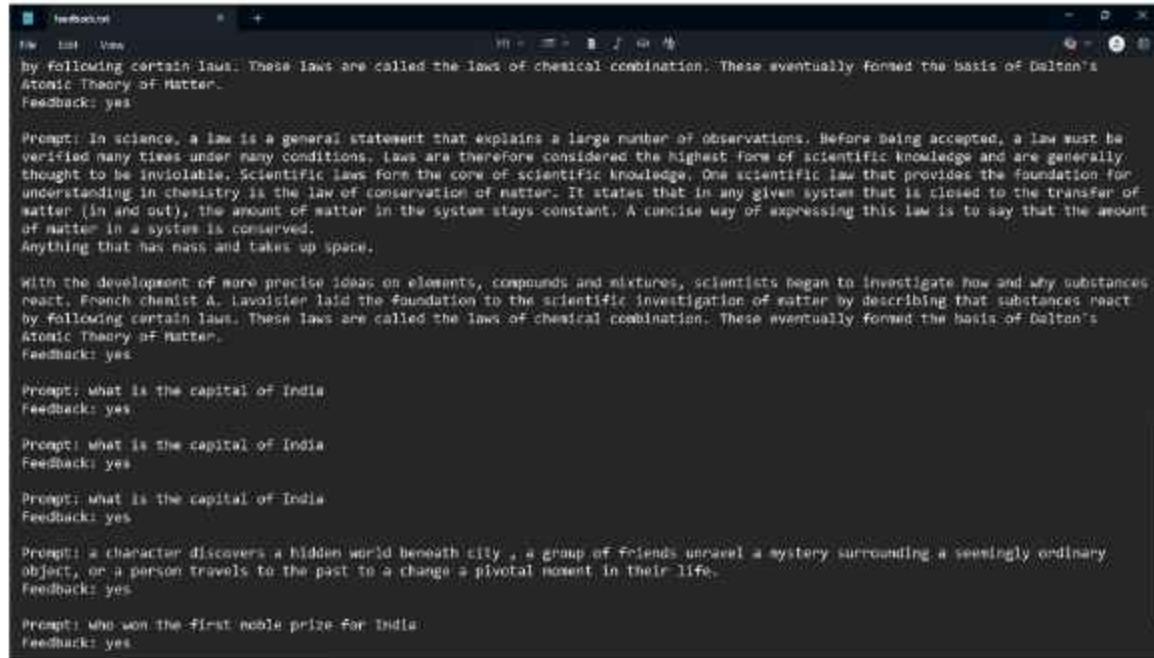
🎨 Generate creative content

Your prompt or text:

a character discovers a hidden world beneath city , a group of friends unravel a mystery surrounding a seemingly ordinary object, or a person travels to the past to a change a pivotal moment in their life.

🔥 Ask the Assistant

7. Feedback Log Example



```
feedback.txt
File Edit View
by following certain laws. These laws are called the laws of chemical combination. These eventually formed the basis of Dalton's Atomic Theory of Matter.
Feedback: yes

Prompt: In science, a law is a general statement that explains a large number of observations. Before being accepted, a law must be verified many times under many conditions. Laws are therefore considered the highest form of scientific knowledge and are generally thought to be inviolable. Scientific laws form the core of scientific knowledge. One scientific law that provides the foundation for understanding in chemistry is the law of conservation of matter. It states that in any given system that is closed to the transfer of matter (in and out), the amount of matter in the system stays constant. A concise way of expressing this law is to say that the amount of matter in a system is conserved.
Anything that has mass and takes up space.

With the development of more precise ideas on elements, compounds and mixtures, scientists began to investigate how and why substances react. French chemist A. Lavoisier laid the foundation to the scientific investigation of matter by describing that substances react by following certain laws. These laws are called the laws of chemical combination. These eventually formed the basis of Dalton's Atomic Theory of Matter.
Feedback: yes

Prompt: what is the capital of India
Feedback: yes

Prompt: what is the capital of India
Feedback: yes

Prompt: what is the capital of India
Feedback: yes

Prompt: a character discovers a hidden world beneath city , a group of friends unravel a mystery surrounding a seemingly ordinary object, or a person travels to the past to change a pivotal moment in their life.
Feedback: yes

Prompt: who won the first noble prize for India
Feedback: yes
```

8. System Requirements

Hardware Requirements

- Intel i3 processor or equivalent (i5/i7 recommended)
- Minimum 4 GB RAM (8 GB recommended)
- At least 500 MB disk space
- Stable internet connection
- Screen resolution 1366×768 minimum

Software Requirements

- OS: Windows 10+, macOS, or Ubuntu 20.04+
- Python 3.8 or higher
- pip (latest)
- Browser: Chrome, Firefox, Edge, or Safari
- IDE/Text Editor: VS Code, PyCharm (optional)

Python Dependencies

- Flask
- requests
- python-dotenv

API Requirements

- Valid OpenRouter API key
- Store the API key in an environment variable

Other Requirements

- Ability to manage virtual environments
- File system access to log feedback

9. Conclusion

The AI Assistant demonstrates prompt engineering principles, API integration, and user interaction handling. It can be extended to:

- Support more tasks
- Add user authentication
- Provide analytics dashboards